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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations August-2025

Computer Science and Engineering

COMPILER DESIGN

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	What is a language processor? Give one example.	01	1	1
	1.2	Define the main role of a lexical analyzer in a compiler. Give example.	02	1	2
	1.3	Mention the two pointers used in input buffering and their roles.	02	1	2
	1.4	What does LR stand for in LR parsing?	01	1	2
	1.5	E → T E' E' → + T E' | ε T → id Construct the first and follow set for the above grammar.	02	2	2
	1.6	Using shift -reduce parsing show steps to parse the input string "aabb" for grammar: $S \Rightarrow aSb \mid \varepsilon$	02	2	2
	1.7	Given the grammar with the Productions($S \rightarrow YZ$, $S \rightarrow .aSa$, $S \rightarrow .b$, $Y \rightarrow y$, $Z \rightarrow z$} Is there any conflicts in the SLR paring table for the below state? Give reason. If yes, mention the type of conflict in the SLR parsing table. $I10 = X \rightarrow YZ.$ $X \rightarrow .aX$ $X \rightarrow .b$	02	3	2
	1.8	What are the major storage areas used in runtime memory layout?	01	1	3
	1.9	What is the benefit of stack allocation for recursive functions?	01	1	3
	1.10	Draw the directed Acyclic Graph (DAG) for the expression: a+ a* (b-c)+(b-c) * d	02	3	3
	1.11	Write the type expression for: int arr[5][10]	01	2	4
	1.12	Give intermediate code for calling a function sum(x,y)	01	2	3
	1.13	Define 2 important design issues in code generation.	02	1	3

PART-B

2	a	Explain lexical analysis, its importance and different methods used in lexical analysis. Provide examples to illustrate each method.	08	2	2
	b	Construct the transition diagram that recognizes the lexemes matching the token relop (relation operator) and explain. Illustrate the implementation of relop transition diagram.	04	3	4
	c	Write down the output of each phase of compiler for the expression a:= b+c * 50.	04	3	2

3	a	Construct predictive parser for the following Grammar S → (L) | a L → L,S | S and parse the input string (a,(a,a))	06	3	2
	b	Construct SLR (1) parsing table for the following grammar E → E+T | T T → TF | F F → F * | a | b. Show the acceptance for the string "a+b *a"	10	3	2
		OR			
4	a	Check whether the grammar is LALR (1) but not SLR(1) S → Aa | bAc | dc | bda A → d	06	3	2
	b	Explain shift-reduce parsing and construct a parse stack trace for the string "aaaab" using the given grammar. S → AA A → aA A → b	04	3	2
	c	What are the sources and types of parsing conflicts in LR parsing? Give examples.	06	1	3
5	a	Write a syntax directed definition for evaluating arithmetic expressions involving +,* and integers.	04	3	4
	b	Construct a syntax-directed definition to convert a binary number into its decimal representation. The grammar for generating decimal number is as follows: S → L L → LB | B B → 0 | 1 Show the annotated parse tree for the binary number "1010" .	08	4	4
	c	Explain the structure of an activation record and describe the purpose of each field.	04	1	3
		OR			
6	a	Describe how stack allocation works for managing run time memory. Illustrate with a function call example.	08	2	4
	b	Compare static and dynamic memory allocation. How does heap memory management support dynamic allocation?	08	2	4
7	a	Translate the expression (x+(-y/z) *r)/(t+u) into a sequence of Quadruple, Triple and Indirect triple.	08	3	3
	b	Write down the semantic rules for the below expression and label them using Backpatching technique. Assume the address of the first instruction generated is 100. (a==b) && (c== d && e==f)	08	3	3
		OR			

8	a	Translate the following program to
		i) Three address statements
		ii) Represent it in quadruple
		for i from 1 to 10 do
		for j from 1 to 10 do
		C[i , j] = i*j;
		for i from 1 to 10 do
		C[i , i] = C[i, i] + A [i, i];			

	b	Assume: Dimension of matrix A,C is 10 x 10 matrix entries are numbers that requires 8 bytes and matrix entries are stores in row-major order.	10	4	3
		Construct the DAG for the given statements u=y/z y = u*v x=v+w z=x-u v=v+w	06	4	3
9	a	Identify the leaders, construct the basic block and draw the flow graph for the given three address code. i) i=0 ii) if i>= 10 goto 6 iii) t1= i * i iv) a[i]= t1 v) i=i+1 vi) goto 2	06	4	1
	b	Explain the principal sources of optimization in compiler design with suitable examples.	06	2	3
	c	Differentiate between forward and backward data-flow analysis with examples.	04	1	4
		OR			
10	a	Describe the issues in the code generation with proper explanation and example.	10	1	4
	b	Explain the Code Generation Algorithm in compiler design. Discuss the key seps involved and illustrate with an example.	06	2	3